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Task 4

1. Explain the difference between syntactic pattern recognition (e.g., regex) and statistical pattern recognition (e.g., TF-IDF + classifier) as applied to text. Give one task where each approach alone is clearly insufficient.

Answer: Syntactic pattern recognition works by matching the pattern exactly as it is (i.e., “auto” will match exactly “auto”), it uses exact parsing rules to see if the data matches the predetermined syntax and as such it is sensitive to typos. While statistical pattern recognition relies on probability and data features instead (i.e., given n examples, there’s p probability that this object is “auto”), it uses decision boundaries to decide and classify. An example where each approach alone is not sufficient is when we try to extract information out of historical records from multiple sources, syntactic approach alone will fail because of the very high variations in the formatting, while statistical approach alone is not enough because it lacks structural context, like it may read incorrectly read 13/32/2001 (even though it is nonsensical) as valid date simply because it thinks it is very likely a date line.

2. Bag-of-words and TF-IDF both discard word order. Describe a short example sentence pair where this loss of order changes the meaning, and name one representation covered in class (or from outside reading) that would preserve it.

Answer: “Six stormchasers chase two storms” with “Six storms chase two stormchaser”. Even though to us the two sentences clearly have different meanings, to BoW and TF-IDF those two are the same collection of words [chase, Six, stormchaser, storms, two]. However, a sequence based representation such as N-Grams preserve local order like [Six stormchasers chase, stormchasers chase two, chase two storms].

3. The "curse of dimensionality" is often discussed in pattern recognition. How does it manifest specifically in bag-of-words style text representations, and what is one common technique to mitigate it?

Answer: The curse of dimensionality manifests itself in BoW in the form of sparsity, this is because each document contains only a tiny fraction of the vocabulary, so most vector entries are zero, very high dimensionality makes storing and processing computationally expensive. One of the ways to mitigate this is by doing Feature Selection, which is picking a smaller subset of original variables without changing them. Feature Selection includes stopwords removal, TF-IDF based selection, and frequency filtering among many others.

4. Suppose your spam classifier performs very well on your test set but poorly in production. List two properties of the training data (relative to real-world text) that could explain this gap.

Answer:

1. The training data may not be really representative of the real world test, this can be that the training data is outdated so it doesn't cover the latest spam circumvention tactics, or the training data may come from a different source which has a different reality.
2. The training data is not comprehensive enough, if the training data only contains limited examples or a lot of near duplicate examples the classifier may become hyper specialise on only those patterns and thus fail to generalise when exposed to unseen words, templates, and spam circumvention techniques in actual production.